

Desmos Mastery - Day 1

12 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Course Intro | 5 questions

[Lesson video: Course Intro](#)

1. A problem asks for $3x + 2$, where x is the positive solution of $x^2 - 5x - 14 = 0$. Which plan directly answers the question?
 - A) Find both roots, use the positive root, then evaluate $3x + 2$.
 - B) Read the y-intercept of the quadratic.
 - C) Use the smaller root because it appears first.
 - D) Report the positive root without another calculation.
2. A graphing window shows no intersection of $y = 0.02x + 12$ and $y = 0.03x + 2$. Which next step is justified?
 - A) Conclude that the system has no solution.
 - B) Expand the window or solve the equations algebraically.
 - C) Change a coefficient until an intersection appears.
 - D) Round both slopes to zero.
3. A student finds that two expressions have the same value at $x = 0$ and concludes they are equivalent for every x . What would best strengthen the work?
 - A) Check that both expressions have a y-intercept.
 - B) Use another screenshot of the same point.
 - C) Simplify the difference algebraically and check the domains.
 - D) Change every variable to zero.

4. A system of three equations has a point that satisfies the first two equations. What must the student do before accepting that point?

- A) Confirm that it also satisfies the third equation.
- B) Find the average of the three y-intercepts.
- C) Discard the third equation.
- D) Choose the intersection with the greatest x-coordinate.

5. After a wrong answer, which review note would be most useful for preventing the same mistake?

- A) I need to be more careful.
- B) I forgot to test whether the boundary was included; next time I will check $<$ versus $\leq$ before choosing a point.
- C) This question was unfair.
- D) I should memorize this question's answer.

Desmos Intro | 7 questions

[Lesson video: Desmos Intro](#)

6. A quadratic is shown below. What is its minimum value?

$$f(x) = x^2 - 8x + 7$$

Answer: _____

7. The table lists the number of pages read on five days: 12, 18, 18, 25, and 42. What is the median?

Answer: _____

8. For the function below, what is $f(-3)$?

$$f(x) = 2x^2 + x - 1$$

Answer: _____

9. An angle measures 150° . What is its sine?

$$\sin(150^\circ)$$

- A) $-\sqrt{3}/2$
- B) $-1/2$
- C) $1/2$
- D) $\sqrt{3}/2$

10. Points A($-5, 2$) and B($7, 10$) are the endpoints of a segment. What is the y-coordinate of its midpoint?

Answer: _____

11. A jacket costing \$80 is discounted by 15%, then a 6% sales tax is applied to the discounted price. What is the final price, in dollars?

Answer: _____

12. A function is defined only for $1 \leq x < 5$. Which point belongs to its graph?

$$f(x) = 3x + 1 \quad (1 \leq x < 5)$$

- A) (0, 1)
- B) (1, 4)
- C) (5, 16)
- D) (6, 19)

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: A 2: B 3: C 4: A 5: B 6: -9 7: 18 8: 14 9: C 10: 6 11: 72.08 12: B

Why each answer works

1. Answer: A. The roots are -2 and 7 . The positive-root condition selects 7 , and the requested expression is 23 . Identify both the restriction and the target before entering anything.

2. Answer: B. Different slopes ensure one intersection. Solving gives $x = 1,000$ and $y = 32$, outside a typical initial window. A missing visible point is not proof of no solution.

3. Answer: C. One matching value cannot establish an identity. For example, x and x^2 agree at zero but differ at two. An algebraic identity with matching domains establishes equivalence.

4. Answer: A. A solution of a system must satisfy every equation simultaneously. This targets the common-intersection error from your PSAT.

5. Answer: B. The useful note names the missed condition and gives a specific future action. Apply that action to a different problem to check transfer.

6. Answer: -9. Complete the square: $(x - 4)^2 - 9$. The vertex is $(4, -9)$; the minimum is the y -coordinate, not the x -coordinate.

7. Answer: 18. The list is ordered and has five entries. The third entry, 18 , is the median. The mean is 23 and answers a different question.

8. Answer: 14. Use parentheses around the negative input: $2(-3)^2 + (-3) - 1 = 18 - 3 - 1 = 14$.

9. Answer: C. Use degree mode for the stated degree measure. The reference angle is 30° and sine is positive in quadrant II, so the value is $1/2$.

10. Answer: 6. Average corresponding coordinates: $((-5 + 7)/2, (2 + 10)/2) = (1, 6)$.

11. Answer: 72.08. The discounted price is $80(0.85) = 68$. The final price is $68(1.06) = 72.08$. The tax applies to the discounted price.

12. Answer: B. The rule gives $y = 3x + 1$, so $(1, 4)$ satisfies it and includes the left endpoint. The right endpoint $x = 5$ is excluded.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.